

CHAPTER 1 — WHAT AI IS, AND WHY CYBERSECURITY MATTERS

Introduction

This book is about how to limit proliferation of bad AI in the world as this technology works its way into every element of everyday living, from the internet to the IoT to transportation modes. With the rush to monetization we should not sacrifice security for a rush to ROI. However, that is not a realistic situation as shall be seen in this book that reviews current cybersecurity for AI and Humanoid Robots. Other elements that make cybersecurity a primary goal in developing AI is that it also makes AI more efficient and minimizes errors, so cybersecurity is value additive not subtractive to long term growth and sustainable monetization pathways. This work's focus is on Large Language Models (LLMs), AI Agents, and Humanoid Robots.

What AI Is, and Why Cybersecurity Matters

Artificial Intelligence (AI) has existed in various forms for decades, but the arrival of modern large language models (LLMs), multi-modal systems, and tool-using autonomous agents has transformed AI from a specialized research field into a pervasive, general-purpose capability. Unlike traditional software, which follows deterministic rules written explicitly by programmers, modern AI systems **learn patterns from vast amounts of data** and use probabilistic inference to generate text, make decisions, reason, or take actions. This shift is foundational: it means that AI can behave in ways its creators never directly programmed, including ways they can't always predict or fully control.

From a cybersecurity standpoint, this matters profoundly. The systems we are now deploying across society—from customer-service chatbots to autonomous drone fleets—are **not simple software packages** with known, inspectable failure modes. They are statistical reasoning engines embedded into critical infrastructure, financial markets, hospitals, operating systems, and cloud environments. They interact with users, adversaries, code execution tools, APIs, networks, payment systems, and other AIs. In many use cases, they are granted autonomy: the ability to perform multi-step tasks, call tools, write code, control hardware, or operate continuously without direct human supervision.

This combination—unpredictability, autonomy, and integration with critical systems—creates a **new class of cybersecurity challenges**. Traditional cyber defense assumes that vulnerabilities exist in code, protocols, misconfigurations, or social engineering. AI introduces new attack surfaces:

- **Prompt injection**, where attackers manipulate the model through natural language rather than exploiting code.
- **Model poisoning**, where training data is tampered with to insert backdoors or biases.
- **Tool hijacking**, where an AI agent is tricked into executing harmful actions using its authorized capabilities.
- **Model leakage**, where sensitive knowledge is extracted or confidential training data is revealed.

- **Autonomous runaway behaviors**, where an AI continues tasks beyond intended scope.
- **Abuse by users**, who leverage models for fraud, malware, impersonation, or manipulation.

Crime becomes accessible at hyper-scale to novices, AI transforms not only how attacks occur, but **who can carry them out**. Actions that once required expert programmers or nation-state resources can now be executed by individuals with minimal expertise. Fraud, deepfake identity deception, disinformation campaigns, social engineering, and code generation have become accessible to millions. The barrier to entry for cybercrime—and influence operations—has collapsed.

The chapter also introduces why the **distinction between safety and security** is crucial. Safety addresses how the model behaves under normal use (e.g., reducing harmful outputs), while security concerns adversarial misuse and attacks designed to subvert or manipulate the model. These domains overlap but are not identical. Safety guardrails can be bypassed by targeted adversaries; likewise, secure deployment does not guarantee safe behavior.

Finally, this chapter frames the book’s core thesis: **AI cybersecurity is not optional**. As AI becomes embedded into the global digital and physical infrastructure, securing these systems becomes integral to safeguarding society. The goal of this book is to provide developers, cybersecurity professionals, policymakers, and educated non-technical readers with the conceptual and practical tools needed to understand and defend the emerging AI ecosystem. Everything else builds on this foundation.

Researchers have attempted to anticipate threats that may develop from AI, for instance from 2018:

Artificial intelligence (AI) and machine learning (ML) are altering the landscape of security risks for citizens, organizations, and states. Malicious use of AI could threaten digital security (e.g. through criminals training machines to hack or socially engineer victims at human or superhuman levels of performance), physical security (e.g. non-state actors weaponizing consumer drones), and political security (e.g. through privacy-eliminating surveillance, profiling, and repression, or through automated and targeted disinformation campaigns).

The malicious use of AI will impact how we construct and manage our digital infrastructure as well as how we design and distribute AI systems, and will likely require policy and other institutional responses. The question this report hopes to answer is: how can we forecast, prevent, and (when necessary) mitigate the harmful effects of malicious uses of AI? (Brundage, 2018)

Yet, even with forward thinking attempts to anticipate attack vectors, we see that those anticipations cannot prognosticate how malicious actors in the real world have used AI to their advantage through emerging attack vectors. In this work some of these attacks shall be covered including countermeasures to these attacks.

As this technology is new the old way of doing cybersecurity that we have grown accustomed to will no longer be relevant. For instance regarding LLMs:

“Large language models (LLMs) have emerged as a transformative force in the rapidly evolving information technology landscape, offering unprecedented capabilities in natural language

processing, content generation, and decision support. Integrating LLMs into enterprise operations is not merely a technological upgrade; it represents a fundamental shift in how organizations process information, interact with customers, and make decisions.

However, as with any emerging technology, adopting LLMs introduces new vulnerabilities and risk factors that must be carefully managed. From data privacy concerns to the potential for malicious manipulation, the security implications of LLM deployment are far-reaching and complex. Organizations must develop comprehensive strategies for secure LLM deployment in enterprise settings.” (Malik, 2024)

Security Concerns for Large Language Models: A Survey



Image: Taxonomy of Threat Surface to LLMs (Li, 2025)

One important concept to know for LLMs is if they are ‘aligned’ or not. To be aligned means that they have adequate guardrails to protect against dangerous content being exploited by bad actors, if they prevent such malicious use the model is known as being ‘aligned’ if it is exploitable for malicious purposes it is ‘non-aligned’. This is also related to the concepts of jailbreaking of a LLM which refers to cracking an ‘aligned’ model to generate ‘non-aligned’ content, thus breaking it’s guard rails.

AI as an Engine of Pattern Recognition

Modern AI systems — particularly large language models (LLMs) — are **predictive engines** trained to identify high-dimensional patterns in data⁴. They do not store facts in databases; instead they learn statistical associations across billions of parameters⁵.

When an AI model generates text, it is computing the most probable next token given its context. This single design principle gives rise to reasoning-like properties — but also to vulnerabilities exploitable by adversaries⁶.

Understanding AI Through Embeddings

One of the elements of AI is the processing of natural language (NLP) where human text is transformed into mathematical representations, that is breaking down the syntax, semantics and memes into maths. Although, one can see the potential problems with this out-of-the box getting a machine to accurately infer human speech through an extrapolation layer of maths it is how engineers get machines to process language. Tokenization is breaking down a larger stream of text into smaller textual units, called tokens, which can be in various forms, from individual characters to full words or phrases.

Tokenization is performed to enhance the model interpretability and ease in processing. Word Embeddings encode meaning as numerical vectors. Words, concepts, and even images are placed into geometric relationships within high-dimensional latent space⁷.

Attackers exploit this by crafting prompts, poison data, or adversarial samples that manipulate these semantic relationships⁸. We shall learn more about these exploits in later chapters.

Read more at: <https://www.geeksforgeeks.org/nlp/tokenization-vs-embeddings/>

Deep Learning Architectures in Plain Language

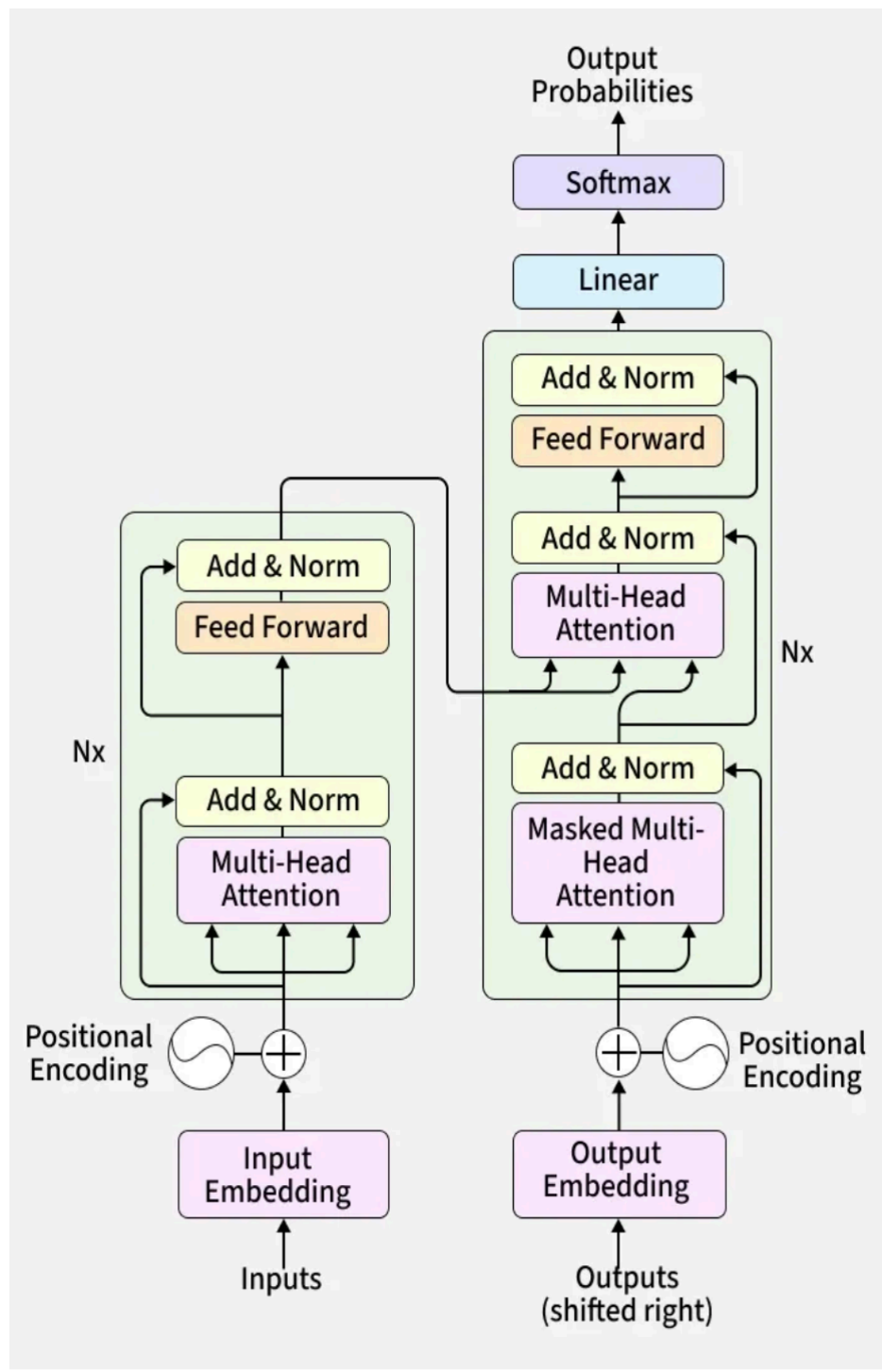
Modern AI systems largely rely on **Transformer** architectures which were introduced in 2017 by Vaswani et al. published a paper "Attention is All You Need" in which the transformers architecture, which use self-attention mechanisms to determine which parts of input data are most relevant⁹.

For cybersecurity:

- Transformers generalize based on patterns, not explicit rules.
- Their internal states are opaque.
- They can behave unpredictably under adversarial inputs¹⁰.

This unpredictability is a central reason why AI security differs dramatically from classical security engineering.

Why LLMs Are Not Traditional Software



1.2.1 No Deterministic Code Paths

Image from: <https://www.geeksforgeeks.org/machine-learning/getting-started-with-transformers/>

Classical software is deterministic: the same inputs yield the same outputs, which is to say that if I program in $2+2 = 4$ it will always return 4 when I call the addition function `add(2,2)`, but in LLMs this

is not the case you can get a different result by repeating the same prompts for instance, even when they are verbatim cut and paste over again into the same LLM.

LLMs are **stochastic**, producing different outputs even with identical prompts¹¹. As a result:

- Vulnerabilities cannot be fixed via a single patch.
- Behavior cannot be fully audited.
- Security flaws emerge from training data and learned patterns rather than code defects¹².

Opaque Internal Reasoning (Reading a Blackbox)

AI “reasoning” occurs within billions of numerical weights, making internal decision processes untraceable¹³.

The remarkable thing is how tractable and meaningful these circuits seem to be as objects of study. When we began looking, we expected to find something quite messy. Instead, we’ve found beautiful rich structures, often with [symmetry](#) to them. Once you understand what features they’re connecting together, the individual floating point number weights in your neural network become meaningful! *You can literally read meaningful algorithms off of the weights.* (Olah, 2022)

This opacity forces cybersecurity teams to rely on **behavioral testing**, **adversarial probing**, and **continuous monitoring**, not code review¹⁴.

Emergent Properties

When models reach sufficient scale, they exhibit emergent abilities — including coding, tool use, and multi-step reasoning — that were **not explicitly programmed**¹⁵.

Emergence is unpredictable, which introduces unique risks such as unintended planning or unsafe autonomous behavior¹⁶. We shall cover the dangers that emergent properties entail in AI in a later chapter.

we introduce generative agents: computational software agents that simulate believable human behavior. Generative agents wake up, cook breakfast, and head to work; artists paint, while authors write; they form opinions, notice each other, and initiate conversations; they remember and reflect on days past as they plan the next day. To enable generative agents, we describe an architecture that extends a large language model to store a complete record of the agent’s experiences using natural language, synthesize those memories over time into higher-level reflections, and retrieve them dynamically to plan behavior. We instantiate generative agents to populate an interactive sandbox

environment inspired by The Sims, where end users can interact with a small town of twenty-five agents using natural language. In an evaluation, these generative agents produce believable individual and emergent social behaviors. (Park, 2023)

We observed evidence of emergent outcomes across all three cases. During the two-day simulation, the number of agents who knew about Sam's mayoral candidacy increased from one (4%) to eight (32%), and the number of agents who knew about Isabella's party increased from one (4%) to thirteen (52%), all without any user intervention. None who claimed to know about this information had hallucinated it. We also observed that the agent community formed new relationships during the simulation, with the network density increasing from 0.167 to 0.74. Out of the 453 agent responses regarding their awareness of other agents, 1.3% (n=6) were found to be hallucinated. Lastly, we found evidence of coordination among the agents for Isabella's party. The day before the event, Isabella spent time inviting guests, gathering materials, and enlisting help to decorate the cafe. On Valentine's Day, five out of the twelve invited agents showed up at Hobbs cafe to join the party. (Park, 2023)

New Attack Surfaces

AI introduces attack vectors that did not previously exist as relayed by MITRE Atlas, which provides guidance on AI Cybersecurity:

- Prompt injection
- Chain-of-thought hijacking
- Self-modifying agent loops
- RAG poisoning
- Model extraction
- Weight theft
- GPU hijacking and covert training¹⁷

These attacks target semantic behavior rather than traditional code vulnerabilities. Using language to hack a language model.

Attackers No Longer Need Expertise

AI dramatically lowers the barrier to cybercrime. One way that cybercriminals can maximize AI for crimes is the adoption of open source LLMs, which are termed 'DarkLLM', not just because they are promoted on the Dark Web, but also because they have evil intentions.

In prior decades, malware development required deep technical skill. Now, a novice can simply ask a DarkLLM for exploit code or phishing templates¹⁸. This accessibility of offensive capability is historically unprecedented.

AI Makes Attacks Faster, Cheaper, and Scalable

AI systems can automate reconnaissance, phishing, exploit generation, and social engineering at global scale¹⁹. DarkLLMs already offer “malware-as-a-dialogue” capabilities, enabling attackers to iterate rapidly²⁰.

Safety = Preventing Harmful Model Output

Safety alignment focuses on harmful outputs: bias, toxic content, dangerous recommendations, and self-harm content²¹.

Security = Preventing Attacks on the Model

Security protects the **model itself** — its inputs, internal state, and tool access — from manipulation²².

You can have a safe-but-insecure model (easy to attack) or a secure-but-unsafe model (harmful outputs).

Both must be addressed simultaneously.

Case Studies

Case Study 1: LLM-Driven Corporate Data Breach Companies inadvertently leaked sensitive documents into LLM memory buffers used for training, later extractable via indirect jailbreaking²³.

Case Study 2: Prompt Injection in Autonomous Banking Bot An email containing hidden adversarial instructions manipulated an AI assistant into unauthorized API calls²⁴. This is an example of cross-domain prompt injection via untrusted content channels.

Case Study 3: DarkLLM-Assisted Malware Operations Cybercriminals used unaligned LLMs to generate polymorphic malware that evaded signature-based defenses by mutating code every few minutes²⁵.

Takeaways:

- Modern AI systems operate on statistical inference, not explicit logic.
- Their internal reasoning is opaque and emergent.
- AI introduces novel attack surfaces, many targeting semantics rather than code, expanding attack vectors.
- The accessibility of AI capability accelerates global cyber risk.
- Securing AI requires **new paradigms** beyond classical cybersecurity.

Footnotes

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